Cosmological Redshift in FFGFT

A Brief Summary for the IPI Correspondence

Johann Pascher

ORCID: 0009-0000-6518-4064

GitHub: https://github.com/jpascher/T0-Time-Mass-Duality

Zenodo (FFGFT v1.0.13): https://zenodo.org/records/20041529

DOI: 10.5281/zenodo.20041529

May 2026

Abstract

In FFGFT, cosmological redshift is a geometric path effect, not a Doppler signature of a metric expansion. Photons traverse the fluctuating ξ-vacuum along geodesics whose effective path length L_{eff} is systematically longer than the Euclidean distance d, with $z = (L_{\text{eff}} - d)/d$. The Hubble constant becomes a fundamental coupling that follows from the single dimensionless $\xi = 4/30{,}000$:

$$\frac{\hbar H_0}{m_e c^2} = \frac{\pi}{2}\,\xi^{10} \qquad \Longrightarrow \qquad H_0 \approx 66{,}5 \text{ km/s/Mpc},$$

parameter-free, within $1{,}3\%$ of the Planck value. This note summarises the FFGFT redshift mechanism, the two equivalent derivations of H_0, and the natural resolution of the Hubble tension. Source documents in the FFGFT corpus: Doc 025 (cosmology), Doc 026 (geometric cosmology), Doc 027 (Solar System consistency), Doc 165 (H_0 ratio), Doc 166 (Casimir-Hubble bridge).

1 Setting

FFGFT models the universe as static and flat. The single dimensionless parameter $\xi = 4/30{,}000$ governs the granularity of the ξ-vacuum. Photons crossing the vacuum are not carried by an expanding metric but propagate along geodesics that incorporate the small-scale fluctuations of the ξ-field. The cumulative effect over cosmological distances is a systematic redshift; over Solar-System distances it is operationally negligible.

2 Redshift as a Geometric Path Effect

The central observation, formalised in Doc 026 via a finite-element simulation of the ξ-vacuum, is that geodesics through the granular ξ-field are not perfect straight lines. The effective path length L_{eff} is systematically larger than the direct Euclidean distance d between source and observer:

$$z = \frac{L_{\text{eff}} - d}{d}\,, \tag{1}$$

with the linearised approximation

$$z \approx d \cdot C \cdot \xi \tag{2}$$

where C is a geometric constant and ξ controls the cumulative path-stretching rate. The redshift is a property of the spacetime geometry itself, not of the relative velocity between source and observer.

Frequency-independence as a structural test. A geodesic is a property of the geometry; it does not depend on the photon's frequency. A red and a blue photon emitted from the same source follow the *same* extended path. Their wavelengths are stretched by the same proportional factor. This is in agreement with the observed frequency-independence of cosmological redshift – a feature that conventional "tired light" models fail to reproduce, and that distinguishes the FFGFT geometric mechanism from energy-loss scenarios with a frequency-dependent cross section.

3 The Hubble Constant as a Fundamental ξ-Ratio

Two equivalent FFGFT derivations of H_0 are recorded in the corpus.

Doc 165 – the dimensionless ratio. The cleanest formulation expresses H_0 as a pure ratio between the cosmological and the electron rest-energy scale:

$$\boxed{\frac{\hbar H_0}{m_e c^2} = \frac{\pi}{2} \cdot \xi^{10}} \tag{3}$$

Numerically:

$$H_0^{\text{FFGFT}} \approx 66{,}5 \text{ km/s/Mpc} \qquad (\text{Planck: } 67{,}4 \pm 0{,}5, \text{ deviation } +1{,}3\%). \tag{4}$$

The integer exponent 10 reflects how far the cosmological energy scale lies from the lepton mass scale, measured in ξ-units. The geometric prefactor $\pi/2$ originates from the toroidal topology of the ξ-time field.

Doc 026 – the Hubble-energy form. Equivalently, the Hubble energy is expressed as

$$E_H = E_0 \cdot \xi^{41/4} = 1{,}41 \times 10^{-33} \text{ eV}, \qquad H_0 = \frac{E_H}{\hbar} \approx 66{,}2 \text{ km/s/Mpc}, \tag{5}$$

with $E_0 = 7{,}398$ MeV the FFGFT reference energy. The exponent decomposes as

$$\frac{41}{4} = \underbrace{10}_{\text{Doc 165 } (H_0 \text{ ratio})} + \underbrace{\frac{1}{4}}_{\text{Doc 091 (Casimir scale)}}, \tag{6}$$

linking the Hubble derivation to the Casimir-CMB derivation through one and the same ξ.

Reinterpretation. Both forms agree to better than 1% with each other and with the Planck-scale measurement. In FFGFT, H_0 is not an expansion rate. It is a fundamental coupling describing the strength of the photon-vacuum interaction, derived from ξ alone.

4 The Hubble Tension

Standard cosmology faces a $\sim 9\%$ discrepancy between early-universe ($H_0 \approx 67{,}4$ km/s/Mpc, Planck CMB) and late-universe ($H_0 \approx 73$ km/s/Mpc, SH0ES distance ladder; H0LiCOW lensing) measurements. Within ΛCDM this is a structural problem because the expansion rate should be a single universal constant.

In FFGFT the tension dissolves at the conceptual level for three reasons.

1. **No expansion, no expansion rate.** H_0 is not the rate at which space expands. It quantifies the cumulative photon-vacuum interaction. Different measurement methods need not return identical values, and the disagreement between early-universe and late-universe inferences is not by itself a contradiction.

2. **Method-dependent path probes.** Different probes (CMB acoustic peaks, distance-ladder Cepheids, gravitational lensing time delays) sample different distances and different photon energy regimes. In a granular ξ-vacuum, the cumulative coupling effective in each regime can differ systematically. The pattern of early-low / late-high inferred values is what one would expect if ξ-coupling depends mildly on the integrated path properties.

3. **Anisotropy compatibility.** Finite-element simulations of the ξ-vacuum (Doc 026) yield small directional variations of the effective coupling, consistent with reported indications of large-scale anisotropy in H_0 measurements.

5 Consistency with Solar-System Data

A natural concern with any modification of cosmological redshift is its compatibility with high-precision Solar-System tests. FFGFT addresses this directly (Doc 027): the geometric path stretching is a cumulative effect over cosmological distances. Over Solar-System distances (light-hours), the stretching is operationally negligible, and FFGFT predicts *zero* measurable orbital anomalies in the Solar System.

This is a structural difference from modified-gravity approaches to the Hubble tension, several of which were excluded by Nathan et al. (2024, MNRAS 544, 975) on Solar-System grounds. FFGFT does not modify gravity at Solar-System scales; it reinterprets the cosmological-redshift premise. The Cassini-mission constraints, which exclude many modified-gravity proposals, are therefore automatically satisfied.

6 Summary of Predictions

1. Cosmological redshift is a geometric path effect, $z = (L_{\text{eff}} - d)/d$, frequency-independent by construction.

2. H_0 follows parameter-free from $\xi = 4/30{,}000$:

$$\hbar H_0/(m_e c^2) = (\pi/2)\,\xi^{10} \quad \Longleftrightarrow \quad E_H = E_0\,\xi^{41/4}\,,$$

   yielding $H_0 \approx 66{,}2$ – $66{,}5$ km/s/Mpc.

3. The Hubble-tension pattern (early-low / late-high) is structurally compatible with FFGFT and does not require additional model components.

4. The Solar-System Cassini constraints are satisfied automatically; no modification of gravity at local scales.

5. The Casimir scale L_ξ and the Hubble constant H_0 satisfy an algebraic bridge relation, $L_\xi \cdot H_0/c \propto \xi^{41/4}$ (Doc 166).

Document references

Doc 025 *T0 Cosmology.* Static, eternal universe; CMB as ξ-field manifestation; three alternative interpretations of the redshift mechanism (energy loss, cumulative gravitation, spacetime geometry).

Doc 026 *Geometric cosmology.* Finite-element simulation of the ξ-vacuum; redshift as path stretching; derivation of $E_H = E_0\,\xi^{41/4}$; three-level resolution of the Hubble tension.

Doc 027 *MNRAS analysis.* Compatibility with Solar-System tests; comparison with Nathan et al. (2024) on modified gravity.

Doc 165 *The Hubble ratio.* Derivation of $\hbar H_0/(m_e c^2) = (\pi/2)\,\xi^{10}$ with $H_0 \approx 66{,}5$ km/s/Mpc, parameter-free.

Doc 166 *Casimir-Hubble bridge.* The decomposition $41/4 = 10 + 1/4$ and the algebraic relation $L_\xi \cdot H_0/c$.

All documents available at `https://github.com/jpascher/T0-Time-Mass-Duality` and as Zenodo release v1.0.13: `https://zenodo.org/records/20041529`.